Speech Processing

1. a) i) Describe the process of closed-phase LPC analysis. Include appropriate diagrams and relevant equations. [2]

ii) Explain the process of pre-emphasis in the context of LPC analysis. State the main spectral characteristics of signals in the source-filter speech production model, and the effect of pre-emphasis on these signals. [2]

iii) Explain how pre-emphasis should be used in closed-phase LPC analysis. Include justifications for your explanation. [2]

Solution:

(i) The process of closed-phase LPC analysis:

- limits the speech data employed in the estimation of the prediction coefficients to the periods when the glottis is closed.
- uses more than one cycle of speech typically in order to obtain an estimate of the prediction coefficients with sufficiently low estimation variance.
- cannot use more than approx. 3 cycles because of the requirement of spectral stationarity.
- the method to obtain the prediction coefficients is chosen to be the covariance method because it operates with much shorter data lengths than the autocorrelation method.

(ii) Pre-emphasis is used to improve the conditioning of the matrix inversion in estimation of the prediction coefficients.

Variable	Description	Spectral Envelope
$u(n)$	Glottal volume velocity.	~ –12 dB/oct
$u'(n)$	Derivative of glottal volume velocity / voice source signal.	~ –6 dB/oct
$u''(n)$	2nd derivative of glottal volume velocity / derivative of voice source signal / whitened voice source signal.	Flat
$s'(n)$	Preemphasised speech signal / whitened speech signal.	$\sim V(e^{j\omega})$
$s(n)$	Speech signal.	$\sim V(e^{j\omega})$ with -6dB/oct tilt

(iii) No pre-emphasis is required in closed-phase LPC because the excitation spectral tilt normally seen is not present in the closed-phase periods of the glottis.

b) Consider a speech coder employing Code Excited Linear Prediction. The coder uses 10th order autocorrelation LPC analysis.

i) Define Line Spectral Frequencies (LSFs) including appropriate formulae for calculation. [2]

ii) State the advantages of using LSFs in the speech coder under consideration. [2]

iii) Give 3 reasons that bandwidth expansion is used in coders of this type. [2]

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iv) Design a bandwidth expansion scheme such that a pole originally at radius 0.994 has a bandwidth after expansion of 20 Hz. Assume the coder operates with a sampling frequency of 8 kHz.

[4]

Solution:
i) Consider the predictor polynomial $A(z) = 1 - \sum_{j=1}^{p} a_j z^{-j}$. Then form the symmetric and antisymmetric polynomials

$$P(z) = A(z) + z^{-(p+1)} A^*(z^{*-1})$$
$$Q(z) = A(z) - z^{-(p+1)} A^*(z^{*-1})$$

Now define the roots of $P(z)$ to be at $e^{j2\pi f_i}$ for $i = 1,3,5\ldots$
Also define the roots of $Q(z)$ to be at $e^{j2\pi f_i}$ for $i = 0,2,4\ldots$
Then the LSFs are the frequencies $f_1, f_2, \ldots, f_p$.
For full marks it must be stated that $f_0 = 1$ and $f_{p+1} = -1$ always.
ii) Stability check easy. Can interpolate. Vary smoothly in time. Strongly correlated implies better coding. Related to spectral peaks (formants).
iii) Improves LSP quantization; Reduces parameter sensitivity of spectrum; Avoids chirps due to very sharp formant peaks.
iv) 20 Hz unnormalized corresponds to 0.0025, meaning that the required new radius r is given by $(1-r)/\pi = 0.0025$ so that $r = 0.9921$.
The expansion factor k is then given by $k = 0.9921/0.994 = 0.9981$. After expansion the bw is given by

$$b = (1-kr)/\pi = (1-0.9981.0.994)/\pi = 0.0025$$

as required. The scheme finally is to form a new vocal tract filter $V'(z) = V(z/0.9981)$ by multiplying coefficients a_i and b_i by k^i for $k = 0.9981$.

c) Consider a differential speech coding scheme in which $\hat{x}(n)$ represents the signal after quantization such that

$$\hat{x}(n-1) = x(n-1) + e(n-1)$$

where $e(n)$ represents the quantization noise, and the difference signal $d(n)$ is formed as

$$d(n) = x(n) - a\hat{x}(n-1)$$

with constant coefficient a. Show that the variance of $d(n)$ can be written in the form

$$\sigma_d^2 = P\sigma_x^2 + Q\sigma_e^2$$

and find expressions for P and Q.

You may use

$$\rho(i) = \frac{\phi(i)}{\sigma_x^2}$$

where $\phi(i)$ is the autocorrelation function of $x(n)$ and $\rho(i)$ is the corresponding normalized autocorrelation.

[4]

Solution:

$$\begin{aligned}\sigma_d^2 &= E\{d^2(n)\} - E^2\{d(n)\} = E\{d^2(n)\} \\ &= E\{x^2(n) - 2ax(n)\hat{x}(n-1) + a^2\hat{x}^2(n-1)\}\end{aligned}$$

Given

$$\sigma_d^2 = E\{x^2(n)\} - 2aE\{x(n)(x(n-1)+e(n-1))\} + a^2E\{(x(n-1)+e(n-1))^2\}.$$

Assuming the signal and quantization noise are uncorrelated we can write

$$E\{x(n)e(n-1)\} = 0$$

leading to

$$\sigma_d^2 = \left[1 - 2a\rho_x(1) + a^2\right]\sigma_x^2 + \left[a^2\right]\sigma_e^2$$

2. a) Consider a simplified speech production model with the vocal tract filter represented as

$$V(z) = \frac{1 - 2e^{-aT}\cos(bT) + e^{-2aT}}{1 - 2e^{-aT}\cos(bT)z^{-1} + e^{-2aT}z^{-2}}$$

for constants a, b and T.

i) Find the zeros and poles of $V(z)$ and plot the zeros and poles on a labelled diagram of the z-plane. [2]

ii) Find an expression for the impulse response of the simplified vocal tract filter $V(z)$ in terms of $\sin(bT(n+1))$. Consider using partial fraction expansion and the standard z-transform

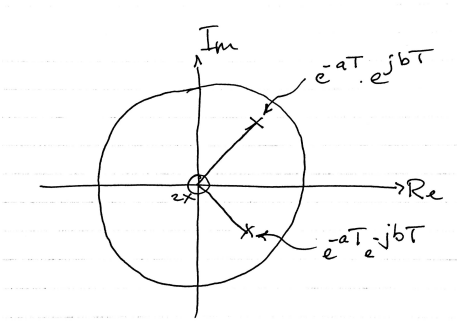
$$\alpha^n u(n) \overset{\text{z}}{\longleftrightarrow} \frac{1}{1-\alpha z^{-1}} \quad |z| > |\alpha|.$$

[4]

iii) Draw a labelled sketch of the impulse response for the case $T = 10^{-4}$, $b = 1000\pi$, $a = 200\pi$ and give the amplitude values of the impulse response at $n = 0$, $n = 3$ and $n = 12$. [2]

iv) Draw a labelled sketch of the magnitude response of the vocal tract filter in dB, using the parameters from Part iii). [2]

Solution:
(i) 2 zeros at $z = 0$, poles at $z = e^{-(a \pm jb)T}$



(ii)

$$V(z) = \frac{A}{1 - d_1 z^{-1}} + \frac{B}{1 - d_2 z^{-1}}$$

for $d_1 = e^{-(a+jb)T}$ and $d_2 = e^{-(a-jb)T}$. Solving for A and B gives

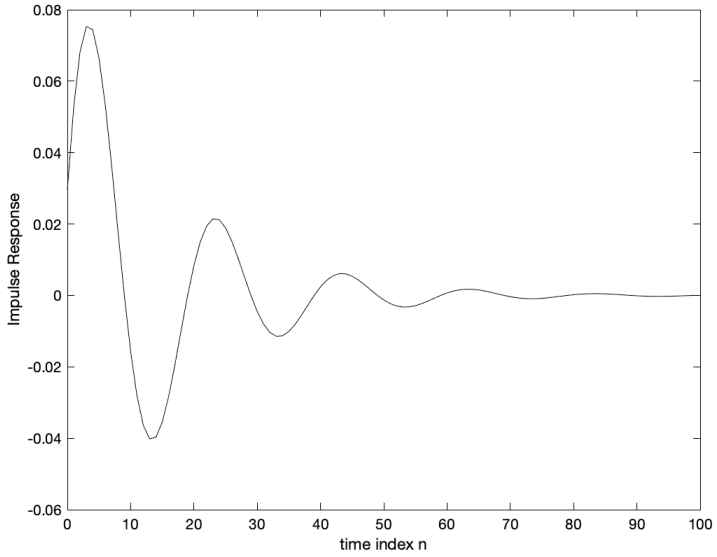
$$A = \frac{1 - 2e^{-aT}\cos(bT) + e^{-2aT}}{1 - e^{j2bT}}$$

$$B = (1 - 2e^{-aT}\cos(bT) + e^{-2aT})\left(1 - \frac{1}{1 - e^{j2bT}}\right)$$

The inverse z-transform is then given by

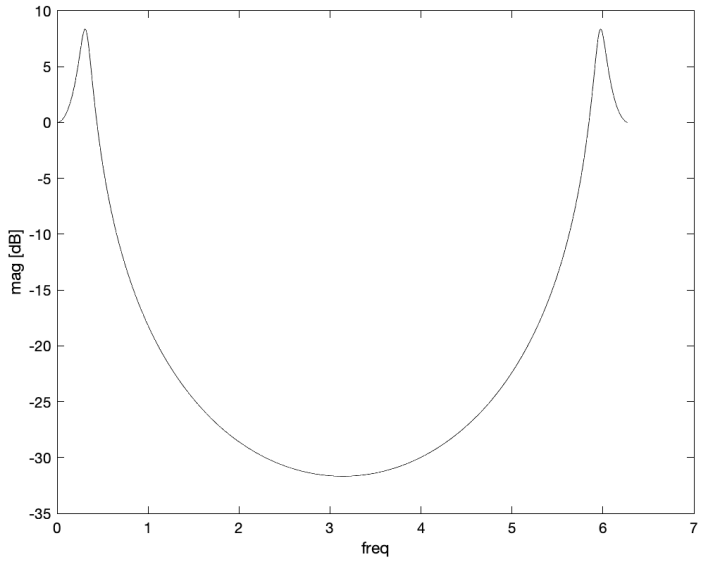
$$\begin{aligned}
v(n) &= \frac{(1 - 2e^{-aT}\cos(bT) + e^{-2aT})}{1 - e^{j2bT}}.e^{-(a+jb)Tn} \\
&\quad + (1 - 2e^{-aT}\cos(bT) + e^{-2aT})\left(1 - \frac{1}{1 - e^{j2bT}}\right) \\
&= \frac{1}{e^{jbT}.2j.\sin(bT)}\Big[e^{j2bT}(1 - 2e^{-aT}\cos(bT) + e^{-2aT}).e^{-(a-jb)Tn} \\
&\quad - (1 - 2e^{-aT}\cos(bT) + e^{-2aT}).e^{-(a+jb)Tn}\Big] \\
&= \frac{(1 - 2e^{-aT}\cos(bT) + e^{-2aT})}{2j\sin(bT)}\left[e^{jbT}.e^{-(a-jb)Tn} - e^{-jbT}.e^{-(a-jb)Tn}\right] \\
&= \frac{(1 - 2e^{-aT}\cos(bT) + e^{-2aT})}{2j\sin(bT)}\left[e^{-aTn}(e^{jbT(n+1)} - e^{-jbT(n+1)})\right] \\
&= \frac{(1 - 2e^{-aT}\cos(bT) + e^{-2aT})}{\sin(bT)}.e^{-aTn}.\sin(bT(n+1)).
\end{aligned}$$

Solution:
(iii)



For $n = [0, 3, 12]$, $v(n) = [0.0296,\ 0.0753,\ -0.0364]$.

Solution:
(iv)



b) Consider a linear prediction filter of the form

$$A(z) = 1 - \sum_{k=1}^{p} a_k z^{-k}$$

for the case of $p = 3$.

i) Find expressions for the cepstral coefficients, c_k, in terms of the prediction coefficients a_k. [2]

Solution:

$$c_1 = a_1$$
$$c_2 = a_2 + \frac{1}{2}c_1 a_1 = a_2 + \frac{1}{2}a_1^2$$
$$c_3 = a_3 + \frac{1}{3}(2c_2 a_1 + c_1 a_2) = a_3 + a_1 a_2 + \frac{a_1^3}{3}$$

ii) Find an expression in terms of the prediction coefficients a_k for the gain in dB of the filter $V(z) = \dfrac{1}{A(z)}$ at a normalized frequency of $\pi/3$. [5]

Solution:
Gain is given by

$$\begin{aligned}
&-10\log_{10}\Big((1-a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3})(1 - a_1 z^{1} - a_2 z^{2} - a_3 z^{3})\Big) \\
&= -10\log_{10}\Big(1 - a_1(z+z^{-1}) - a_2(z^2+z^{-2}) - a_3(z^3+z^{-3}) + a_1^2 \\
&\quad + a_1 a_2(z+z^{-1}) + a_1 a_3(z^2+z^{-2}) + a_2^2 + a_2 a_3(z+z^{-1}) + a_3^2\Big)
\end{aligned}$$

At frequency $\pi/3$ we have

$$(z+z^{-1}) = 1, \quad (z^2+z^{-2}) = -1, \quad (z^3+z^{-3}) = -2$$

and hence

$$\text{Gain } = -10\log_{10}\Big(1 + a_1^2 + a_2^2 + a_3^2 - a_1 + a_2 + 2a_3 + a_1 a_2 - a_1 a_3 + a_2 a_3\Big) \text{ dB}$$

iii) Consider that the prediction coefficients a_k are to be quantized in a speech codec. Using your expression for the gain from your solutions to Part ii), or otherwise, discuss how you would design the quantization scheme for the prediction coefficients for best encoding at a normalized frequency of $\pi/3$, given that the bit-rate is limited. [3]

Solution:
It can be seen that the sensitivity of the gain to all the prediction coefficients is dependent on the square of each coefficient plus a supplementary term. Whereas the sensitivity is equally dependent on the square term for all coefficients, $a3$ has the largest contribution to the supplementary term assuming $a_k < 1, \forall k$. Accordingly, the number of bits should be allocated uniformly to a_1, a_2, a_3, or a_3 could be given slightly more precision (both answers score full marks).

3. a) State and explain the advantages and disadvantages of applying single-channel speech enhancement to speech signals in

i) telephony applications,

ii) speech recognition applications. [6]

Solution:
i) Both uplink and downlink speech enhancement improves perceived audio quality. Removing much of the noise prior to encoding with a speech coder improves coding accuracy for a given bit rate. The enhancement has to be carefully tuned to avoid introducing disturbing artefacts into the sound such as musical noise.
ii) Speech enhancement is not usually successful in improving speech recognition performance because it may modify the speech in a way not seen in the training data. It is normally considered better to perform multicondition training on noisy data.

b) Consider a 'smart home' audio device used for speech input. The device has two microphones separated by 20 cm. The signals at the two microphones are

$$x_1(n) = s(n) + e_1(n)$$
$$x_2(n) = s(n-\delta) + e_2(n)$$

where $s(n)$ is the source speech signal at sample index n, $e_1(n)$ and $e_2(n)$ are independent and zero mean noise signals, and δ describes a time-shift (expressed in samples) required to time-align the component $s(n)$ in both microphone signals.

i) Assuming the talker is at the same height as the microphones and at 2.5 m distance, determine the angle of the talker measured relative to the plane of the microphones for the case $\delta = 472\,\mu$s. Use a sketch to clarify your answer and any assumptions you make. Assume that sound waves propagate at 343 m/s. [4]

Solution:
The path difference between x_1 and x_2 is $0.2\cos(\theta)$ for angle of arrival θ. Converting this distance to time gives

$$\delta = 0.2\cos(\theta)/343$$

so that θ is $\pi/5$. A sketch must be provided to indicate clearly how the angle is measured.

ii) The two microphone signals $x_1(n)$ and $x_2(n)$ are summed to give

$$y(n) = ax_1(n) + (1-a)x_2(n)$$

where a is a chosen real constant.

Find an expression for the variance of the noise contained in the signal $y(n)$. Assume that δ is known. State any assumptions and clearly define any symbols used. [3]

Solution:

$$\begin{aligned}\sigma_3^2 &= E\left[(ae_1(n)+(1-a)e_2(n))^2\right] \\ &= a^2E[e_1^2(n)]+(1-a)^2E[e_2^2(n)] \\ &= a^2\sigma_1^2+(1-a)^2\sigma_2^2\end{aligned}$$

iii) For the same signal $y(n)$, find an expression for the value of a that minimizes the variance of the noise in $y(n)$. [4]

Solution:

$$\frac{\partial\sigma_3^2}{\partial a} = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0$$

$$\Longrightarrow a_{opt} = \frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$$

c) Explain what is meant by the term 'musical noise' in the context of speech enhancement. How can musical noise be minimized in speech enhancement algorithms? Include any relevant mathematical expressions. [3]

Solution:
Random narrowband fluctuations in the residual noise, which are known as musical tones, are annoying and disturbing to the perception of the enhanced signal. Musical noise is typically controlled using a lower-bound, H_{min} on the filter to minimize the noise suppression and decrease speech distortion such as

$$H(l,k) = \max(H(l,k), H_{min})$$

4. a) Consider the following utterances spoken by a female speaker and contaminated by babble noise such that the SNR is −1 dB.

Utterance	Spoken Text
A	"It's a pity to recognize speech badly."
B	"Oh two oh, seven five nine four, six two three five."
C	"Hi Jim, it's Mum. Why don't you answer your phone? Call you later."

i) Write down the rank order of the utterances according to the expected recognition accuracy for a typical automatic speech recognition system. [2]

ii) Explain your choice of rank order. [2]

iii) How would the recognition accuracy, and the rank order you have written down, change if the speech signal were to contain significant reverberation in addition to the babble noise? [2]

iv) Rank order the utterances as in Part i) but now for a human listener instead of an automatic speech recognition system, and explain your choice. [2]

Solution:
i) Rank order C, B, A.
ii) High level of noise means errors will be increased. Babble noise is one of the more difficult noise types. A is very difficult because the string of phonemes 'recognize speech' are ambiguous with 'wreck a nice beach' and the result will likely be determined not by the acoustic information but by the language model. B is difficult because no language modelling can be helpful since the ordering of the digits is unpredictable (expect the first 'oh' for a phone number). However, digit recognition is intrinsically reliable because it is a 'choose one of ten' classifier and, if it can be determined that this string is a phone number, this will help improve accuracy. C is a common phone message that would likely be well modelled in a language model (except for the proper name).
iii) Reverberation will reduce accuracy significantly, commonly up to +100% relative word error rate. No expectation of a change in rank order.
iv) Rank orders C, A, B or C, B, A are both acceptable depending on the explanation given. The explanations are: the semantic context may be used by a human to choose correctly between 'recognize speech' and 'wreck a nice beach'; the rhythmic stress on the important syllables in the phone number may help the listener to recognize the digits more accurately in noise.

b) Compare and contrast the follow approaches to automatic speech recognition:

- automatic speech recognition systems based on neural networks,
- automatic speech recognition systems based on hidden Markov models.

Include in your answer descriptions of the key elements of each approach. [4]

Solution:
Both approaches are data-driven (trained). Both approaches require appropriate input features that are discriminative phonetically. HMMs recognize sequences of features that have high probability of acoustic match with the training data and high probability of occurring in such a sequence. NNs may also include modelling of sequence information according to the temporal context and the architecture employed - for example CNN and LSTM. Additional marks are given for details of either approach, relevant diagrams, and appropriate contextual information in the comparison.

c) Consider a speech recognition system using a hidden Markov model with S states in which the feature vectors for T frames are denoted $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$. The term $d_s(\mathbf{x}_t)$ denotes the probability density of generating an observation frame $\mathbf{x}_t$ from state s. The term $a_{i,j}$ denotes the transition probability from state i to state j of the model.

For a particular example with $T = 6$ and $S = 4$, Table 1 shows the values of $d_s(\mathbf{x}_t)$ and Table 2 shows the state transition probabilities. The symbol '=' indicates the model exit state.

	$d_s(\mathbf{x}_t)$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$
	1	0.5	0.4	0.5	0.2	0.1	0.1
State	2	0.4	0.5	0.7	0.6	0.3	0.8
	3	0.2	0.8	0.2	0.8	0.2	0.2
	4	0.5	0.4	0.5	0.2	0.5	0.8

Table 1: Probability Densities

		j				
	$a_{i,j}$	1	2	3	4	=
	1	0.9	0.1	0	0	0
i	2	0	0.5	0.5	0	0
	3	0	0	0.2	0.8	0
	4	0	0	0	0.5	0.5

Table 2: State Transition Probabilities

i) State any structural features of this hidden Markov model.
[1]

ii) Showing all your working, find the total probability of all alignments of the observation feature vectors with the model for which frame 3 is in state 2. Give your answer to 4 significant figures.
[7]

Solution:
(i) This is a left-to-right no-skips model.
(ii) The total probability is obtained from calculating $P(3,2)Q(3,2)$ where

$$P(t,s) = d_s(\mathbf{x}_t) \sum_{k=1}^{S} a_{ks} P(t-1,k)$$

with initialization $P(1,1) = d_1(\mathbf{x}_1)$, and

$$Q(t,s) = \sum_{k=1}^{S} a_{sk} d_k(\mathbf{x}_{t+1} Q(t+1,k)$$

with initialization $Q(T,S) = a_{s=}$.

$$\begin{aligned}
&P(1,1) = 0.5 \\
&P(2,1) = 0.4 \times (0.9 \times 0.5) = 0.18 \\
&P(2,2) = 0.5 \times (0.1 \times 0.5) = 0.025 \\
&P(3,2) = 0.7 \times (0.1 \times 0.18 + 0.5 \times 0.025) = 0.0213 \\
&Q(6,4) = 0.5 \\
&Q(5,4) = 0.5 \times 0.8 \times 0.5 = 0.2 \\
&Q(5,3) = 0.8 \times 0.8 \times 0.5 = 0.32 \\
&Q(4,4) = 0.5 \times 0.5 \times 0.2 = 0.05 \\
&Q(4,3) = 0.2 \times 0.2 \times 0.32 + 0.8 \times 0.5 \times 0.2 = 0.0928 \\
&Q(4,2) = 0.5 \times 0.2 \times 0.32 = 0.032 \\
&Q(3,2) = 0.5 \times 0.6 \times 0.032 + 0.5 \times 0.8 \times 0.0928 = 0.04672
\end{aligned}$$

Hence $P(3,2)Q(3,2) = 0.0009947$.